

4.7 Cauchy-Euler Equations

An equation of the form:

$$ax^2y'' + bxy' + cy = 0$$

for constants a, b, c is called a
Cauchy-Euler equation.

More generally, an n^{th} order Cauchy-Euler equation has the form:

$$a_n x^n y^{(n)} + \dots + a_2 x^2 y'' + a_1 x y' + a_0 y = 0$$

for constants $a_n, \dots, a_2, a_1, a_0$.

Our main focus in this section is the case
 $n = 2$ (2nd order).

The method of solution to solve:

$$ax^2y'' + bxy' + cy = 0$$

is to look for solutions of the form

$$y = x^m$$

for some number m .

We will look for solutions over the interval $(0, \infty)$.

$$y = x^m \Rightarrow y' = mx^{m-1}$$

$$\Rightarrow y'' = m(m-1)x^{m-2}$$

Plug y, y', y'' into the DE
to obtain:

$$ax^2 \left[m(m-1)x^{m-2} \right] + bx \left[mx^{m-1} \right] + c \left[x^m \right] = 0$$

$$\Rightarrow am(m-1)x^m + bm x^m + c x^m = 0$$

$$\Rightarrow (am(m-1) + bm + c) x^m = 0$$

Since this should hold for all x
in $(0, \infty)$, we conclude that

$y = x^m$ is a solution of the DE

whenever m is a root of

$$am(m-1) + bm + c = 0$$

or

$$am^2 + (b-a)m + c = 0$$

This is the auxiliary equation (or characteristic equation)
of the Cauchy-Euler equation.

Let's consider three cases for the roots
of $am^2 + (b-a)m + c = 0$.

Case ① m_1 and m_2 are distinct real roots

$$y_1 = x^{m_1} \text{ and } y_2 = x^{m_2}$$

are two LI solutions.

You can check that $W \neq 0$.

$$\Rightarrow \boxed{y = c_1 x^{m_1} + c_2 x^{m_2}}$$

Case ② m is the only root

$y_1 = x^m$ is a solution

$$\text{By using : } y_2 = y_1 \int \frac{e^{-\int P(x) dx}}{y_1^2} dx$$

We can deduce that:

$$y_2 = x^m \ln x.$$

$$y = k_1 x^{m_1} + k_2 x^{m_2}$$

$$\Rightarrow y = k_1 x^{\alpha + i\beta} + k_2 x^{\alpha - i\beta}$$

$$\Rightarrow y = k_1 x^{\alpha} x^{i\beta} + k_2 x^{\alpha} x^{-i\beta}$$

$$\Rightarrow y = k_1 x^{\alpha} e^{i\beta \ln x} + k_2 x^{\alpha} e^{-i\beta \ln x}$$

Euler's formula: $e^{i\theta} = \cos \theta + i \sin \theta$

$$\Rightarrow y = x^{\alpha} \left[k_1 (\cos(\beta \ln x) + i \sin(\beta \ln x)) \right] + \dots$$

$$\dots k_2 (\cos(-i\beta \ln x) + i \sin(-i\beta \ln x))$$

$\cos(-\theta) = \cos \theta$ and $\sin(-\theta) = -\sin \theta$

$$\Rightarrow y = x^{\alpha} \left[(k_1 + k_2) \cos(\beta \ln x) + i(k_1 - k_2) \sin(\beta \ln x) \right]$$

if we let: $\begin{cases} c_1 = k_1 + k_2 & \text{and} \\ c_2 = i(k_1 - k_2) \end{cases}$

Then: $y_1 = x^m$ and $y_2 = x^m \ln x$
are two LI solutions.

$$\Rightarrow \boxed{y = c_1 x^m + c_2 x^m \ln x}$$

Case ③ $m_1 = \alpha + i\beta$ and $m_2 = \alpha - i\beta$
are complex conjugate roots.

Then:
$$\begin{cases} z_1 = x^{m_1} \\ z_2 = x^{m_2} \end{cases} \quad \text{and}$$

are complex solutions of the DE.

From the superposition principle,

$$y = k_1 z_1 + k_2 z_2 \text{ is a solution}$$

for any constants k_1 and k_2 .

Then :

$$y = x^{\alpha} \left[c_1 \cos(\beta \ln x) + c_2 \sin(\beta \ln x) \right]$$

We can check with a Wronskian that :

$$y_1 = x^{\alpha} \cos(\beta \ln x) \quad \text{and}$$

$$y_2 = x^{\alpha} \sin(\beta \ln x)$$

are L I.

To solve $ax^2y'' + bxy' + cy = 0$,

We find the roots of:

$$am^2 + (b-a)m + c = 0$$

Case ① m_1 and m_2 are distinct real roots

$$y = c_1 x^{m_1} + c_2 x^{m_2}$$

Case ② m is the only root

$$y = c_1 x^m + c_2 x^m \ln x$$

Case ③ $m = \alpha \pm i\beta$ are complex roots

$$y = x^\alpha \left[c_1 \cos(\beta \ln x) + c_2 \sin(\beta \ln x) \right]$$

ex: ① Solve $4x^2 y'' + 4xy' - y = 0$

Cauchy-Euler

roots: $am^2 + (b-a)m + c = 0$

$$\Rightarrow 4m^2 + (4-4)m - 1 = 0$$

$$\Rightarrow 4m^2 = 1$$

$$\Rightarrow m = \pm \frac{1}{2}$$

Two distinct real roots: $\frac{1}{2}$ and $-\frac{1}{2}$.

Then: $y = c_1 x^{\frac{1}{2}} + c_2 x^{-\frac{1}{2}}$

or $y = c_1 \sqrt{x} + \frac{c_2}{\sqrt{x}}$

is the general solution.

ex: ② Solve $x^2 y'' + 5xy' + 4y = 0$

Cauchy-Euler

Roots: $a m^2 + (b-a)m + c = 0$

$$\Rightarrow m^2 + (5-1)m + 4 = 0$$

$$\Rightarrow m^2 + 4m + 4 = 0$$

$$\Rightarrow (m+2)^2 = 0$$

$$\Rightarrow m = -2 \quad (\text{unique root})$$

then: $y = c_1 x^{-2} + c_2 x^{-2} \ln x$

ex: ③ solve $3x^2 y'' + 6xy' + y = 0$

Cauchy-Euler

Roots : $am^2 + (b-a)m + c = 0$

$$\Rightarrow 3m^2 + (6-3)m + 1 = 0$$

$$\Rightarrow 3m^2 + 3m + 1 = 0$$

$$m = \frac{-3 \pm \sqrt{9 - 4(3)(1)}}{2(3)}$$

$$m = \frac{-3 \pm \sqrt{-3}}{6} = \underbrace{-\frac{1}{2}}_{\alpha} \pm i \underbrace{\frac{\sqrt{3}}{6}}_{\beta}$$

$$y = x^{\alpha} \left[c_1 \cos(\beta \ln x) + c_2 \sin(\beta \ln x) \right]$$

$$\Rightarrow y = x^{-1/2} \left[c_1 \cos\left(\frac{\sqrt{3}}{6} \ln x\right) + c_2 \sin\left(\frac{\sqrt{3}}{6} \ln x\right) \right]$$

ex: ④ Solve the IVP

$$\begin{cases} x^2 y'' + 6xy' + 6y = 0 \\ y(1) = 3 \\ y'(1) = 4 \end{cases}$$

Step ① Solve the DE

It is a Cauchy-Euler equation.

Roots : $m^2 + 5m + 6 = 0$

$$(m+2)(m+3) = 0$$

$$m = -2, -3$$

$$y = c_1 x^{-2} + c_2 x^{-3}$$

Step ② Find c_1, c_2

$$y' = -2c_1 x^{-3} - 3c_2 x^{-4}$$

$$\begin{cases} y(1) = 3 \rightarrow c_1 + c_2 = 3 \\ y'(1) = 4 \rightarrow -2c_1 - 3c_2 = 4 \end{cases} \xrightarrow{\text{solve}} \begin{cases} c_1 = 13 \\ c_2 = -10 \end{cases}$$

The IVP solution is: $y = 13x^{-2} - 10x^{-3}$

ex: ⑤ Find the general solution of

$$x^2 y'' - 3x y' + 3y = 8x^4 e^{2x}.$$

Step ① Find y_c

Cauchy-Euler : $m^2 - 4m + 3 = 0$
 $(m-3)(m-1) = 0$
 $m = 3, 1$

$$y_c = c_1 x^3 + c_2 x$$

Step ② Find y_p

Let's use Variation parameters.

$$\left\{ \begin{array}{l} y_1 = x^3 \\ y_2 = x \end{array} \right\} \text{ from } y_c$$

$$\left\{ \begin{array}{l} f(x) = 8x^2 e^{2x} \end{array} \right. \leftarrow \text{from the Standard form}$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} x^3 & x \\ 3x^2 & 1 \end{vmatrix} = -2x^3$$

$$W_1 = \begin{vmatrix} 0 & y_2 \\ f(x) & y_2' \end{vmatrix} = \begin{vmatrix} 0 & x \\ 8x^2 e^{2x} & 1 \end{vmatrix} = -8x^3 e^{2x}$$

$$W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f(x) \end{vmatrix} = \begin{vmatrix} x^3 & 0 \\ 3x^2 & 8x^2 e^{2x} \end{vmatrix} = 8x^5 e^{2x}$$

$$u_1 = \int \frac{W_1}{W} dx = \int 4e^{2x} dx = 2e^{2x}$$

$$u_2 = \int \frac{W_2}{W} dx = \int -4x^2 e^{2x} dx$$

$$u_2 = (-2x^2 + 2x - 1)e^{2x}$$

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$-4x^2$	$\oplus e^{2x}$
$-8x$	$\downarrow \frac{1}{2} e^{2x}$
-8	$\downarrow \frac{1}{4} e^{2x}$
0	$\downarrow \frac{1}{8} e^{2x}$

$$y_p = u_1 y_1 + u_2 y_2$$

$$\Rightarrow y_p = \cancel{2e^{ax}}(x^3) + \cancel{(-2x^2 + 2x - 1)e^{ax}} x$$

$$\Rightarrow y_p = (2x - 1)x e^{ax}$$

The general solution is

$$y = y_c + y_p$$

$$y = c_1 x^3 + c_2 x + (2x - 1)x e^{ax}$$

ex: ⑥ Solve the following IVP.

$$\begin{cases} x^2 y'' - x y' + y = 2x \\ y(1) = 2 \\ y'(1) = 3 \end{cases}$$

step ① Find y_c

Cauchy-Euler

Roots : $m^2 - 2m + 1 = 0$

$$(m-1)^2 = 0$$

$$m = 1 \leftarrow \text{unique root}$$

$$y_c = c_1 x + c_2 x \ln x$$

Step ② Find y_p

Variation of parameters

$$\begin{cases} y_1 = x \\ y_2 = x \ln x \end{cases} \quad \text{from } y_c$$
$$f(x) = \frac{2}{x} \quad \leftarrow \text{from the standard form}$$

$$W = \begin{vmatrix} x & x \ln x \\ 1 & \ln x + 1 \end{vmatrix} = x$$

$$W_1 = \begin{vmatrix} 0 & x \ln x \\ 2/x & \ln x + 1 \end{vmatrix} = -2 \ln x$$

$$W_2 = \begin{vmatrix} x & 0 \\ 1 & 2/x \end{vmatrix} = 2$$

$$u_1 = \int \frac{W_1}{W} dx = \int -\frac{2 \ln x}{x} dx$$

$$\text{let } u = \ln x \rightarrow du = \frac{1}{x} dx$$

$$u_1 = \int -2u du = -u^2 = -(\ln x)^2$$

$$u_2 = \int \frac{W_2}{W} dx = \int \frac{2}{x} dx = 2 \ln x$$

$$y_p = u_1 y_1 + u_2 y_2$$

$$y_p = -x (\ln x)^2 + (2 \ln x) x \ln x$$

$$y_p = x (\ln x)^2$$

The general solution is $y = y_c + y_p$.

Step ③ Find c_1, c_2

$$y = c_1 x + c_2 x \ln x + x (\ln x)^2$$

$$y' = c_1 + c_2 (\ln x + 1) + (\ln x)^2 + 2 \ln x$$

$$y(1) = 2 \rightarrow c_1 + 0 + 0 = 2 \rightarrow c_1 = 2$$

$$y'(1) = 3 \rightarrow c_1 + c_2 + 0 + 0 = 3 \rightarrow c_2 = 1$$

The solution of the IVP is

$$y = 2x + x \ln x + x (\ln x)^2$$